MBAI 5600G – Applied Integrative Analytics

Capstone Project – Milestone 2

Multimodal Financial Crisis Prediction:

Integrating Market Signals with LLM-Based Financial News Sentiment Analysis

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1. Updated Executive Summary

The financial markets show regime dynamics, which current quantitative methods do not necessarily identify until losses have occurred, and which are hidden. This capstone project fills a critical need in the literature of early crisis detection, as it is the first empirical effort to develop a multimodal framework that combines price-based and language-based signals into a single framework. Do sentiment signals based on financial news headlines predict price based regime shifts in the S&P 500, and if so, by how many trading days? To do this, the project uses two analytical pipelines:

(1) A quantitative pipeline using a Hidden Markov Model (HMM) and ARMA-GARCH volatility modelling to label market regimes from OHLCV data and

(2) A natural language processing (NLP) pipeline to apply a transformer model named FinBERT to daily financial news headlines to generate sentiment-based stress measures.

To validate, a thorough literature survey of 30 scientific papers is performed, encompassing 10 papers published in the last three years (2024–2026), which show that both HMM and transformer are well-established research areas but that explicit quantification of the lead-lag relationship and the use of the SHAP rule-based approach for explainability are still under-explored contributions. Validation is done on three historical crisis events – the Global Financial Crisis (2008), the COVID-19 Market Crash (2020), and the Inflation Volatility Shock (Ivy Shock) (2022). There are datasets that are public, less than 1GB and can be used with Google Colab's free tier limitations. Inference latency is addressed using batch processing and result caching for FinBERT.

2. Literature Review

2.1 Literature Review Summary

2.1.1 Hidden Markov Models for Financial Regime Detection

The first use of Hidden Markov Models in financial time series is due to Hamilton [1] in a seminal paper published in Econometrica which laid the theoretical foundation for modelling nonstationary economic series by means of a set of hidden discrete states. Later studies showed that the model is useful in equity market regime detection: Ang and Timmermann [5] have written a thorough review showing that an HMM with two to four states generally outperforms single-regime benchmarks in capturing bull-bear market alternation. Further progress in this direction have been made recently in two directions. Ardia et al. [4] propose a heteroskedastic network model which incorporates the transitions between the HMM regimes into a graph-based connectivity structure, and provide statistically significant early-warning signals for the S&P 500 crises of 2008 and 2020. Shu and Mulvey [6] suggest an allocation strategy based on the state probability of the HMM that can be shown to be a regime-aware strategy, and show that it can improve the Sharpe ratio by 12% compared with static allocation. These studies validate the feasibility of HMM but without taking into account the text signals. Specifically, Shu and Zhu [7] expand the HMM by adding the duration-dependent transition matrices which are based on macroeconomic covariates to model regime persistence. Their findings are important for the present project as they inform the first-order Markov assumption, which underestimates the duration of a crisis state during systemic shocks, as it is used in the validation process of the project against the 2008 and 2020 crises.

2.1.2 GARCH Volatility Modelling in Stress Periods

In financial econometrics, the GARCH(1,1) model by Bollerslev [3] is still the standard model used for conditional volatility estimation. A vast amount of subsequent literature, summarized by Engle [2], shows that the GARCH family of models are found to explain volatility clustering and fat tailed distributions of returns in equity, FX and commodity markets. GARCH conditional variance estimates are used as input features for the HMM, which are continuous estimates of volatility of the process, rather than discrete labels for regimes; this is critical for this project. Recently, Huang and Luo [8] compared the performance of GJR-GARCH, EGARCH, and standard GARCH specifications on the daily returns of the S&P 500 from 2000 to 2023, concluding that the standard GARCH(1,1) specification is a good choice for estimating volatility during crises, but that asymmetric variants provide marginal improvements only for sub-samples of the bear market. This result confirms that the project's default settings of GARCH(1,1) and rolling standard deviation are reasonable.

2.1.3 Financial Sentiment Analysis with Transformer Models

With the release of BERT by Devlin et al. [9], a series of finance domain adapted language models has been created. Araci's FinBERT is an excellent BERT model fine-tuned with 10-K filings, earnings calls and financial news articles, gaining state-of-the-art performance on the Financial

PhraseBank benchmark. Later work by Shobayo et al. [11] shows that FinBERT performs well on headline-level sentiment classification, especially when compared to lexicon-based models (VADER, Loughran-McDonald) in the context of crisis-period text, which has a vocabulary very different from general English.

One of the major developments documented by Fatouros et al. [14] in their 2023 study is that zero-shot sentiment scores generated by ChatGPT outperform traditional NLP methods and are statistically significant in predicting next-day financial stock returns from the post-2020 era. Their study does not, however, test the prediction of regime or lead-lag timing, which is an open question that this project seeks to answer. Extending FinBERT [10] to Chinese financial text, the model class is further validated with cross-lingual generalisability for financial sentiment. Importantly, Fatouros et al. [14] compare FinBERT with GPT-4 across a set of financial sentiment applications in 2023, and conclude that FinBERT near-matches performance at the level of classification using headlines while outperforming GPT-4 at the level of classification of financial sentiment in nuanced analyst reports; furthermore, they find that FinBERT is significantly more efficient—a key factor for this project's deployment constraint of Colab.

2.1.4 Multimodal Fusion of Price and Text Signals

Using quantitative market signals combined with the features extracted from NLP is an emerging area of research. Twitter sentiment and technical indicators are used by Zhang et al. [23] for prediction of intraday returns, resulting in applicability restrictions for daily regime detection. This project is motivated by the empirical evidence of lead-lag analysis presented by Bollen et al. [24] as they show that the aggregate Twitter mood states Granger-cause DJIA movements three days before they occur.

More structurally relevant is the work of Riso and Vacca [18] who incorporate the sentiment of the earnings call with GARCH volatility forecasts into a multimodal prediction framework. They find that text features boost GARCH forecasts by some 8% in terms of RMSE in the earnings windows, but less advantage is obtained during the tranquil periods; that is, sentiment signals seem to be most informative right in the middle of the stress transitions the project focuses on. To capture the relationships among the price series and news articles in a token-level manner, BloombergGPT [22] proposed a cross-attention fusion mechanism, which outperforms other methods on the benchmarks for stock movement prediction. Further establishing this landscape, Saleh and Semaan

[26] demonstrated that combining qualitative, domain-specific sentiment tracking with hard financial metrics significantly enhances predictive classification baselines in high-risk economic conditions. Most recently, Maqui [30] proved that text-based machine learning frameworks extracting signals from corporate financial communications can successfully predict structural systemic crises and macro-level instability before they manifest in market pricing. The architecture is GPU trained but the concept, namely that the temporal alignment between text and price is an essential factor, directly translates to this project's lead-lag quantification methodology.

2.1.5 Explainability in Financial Machine Learning

In addition to the above, SHAP (SHapley Additive exPlanations) [21] offers theoretically motivated feature attributions that are consistent and offer guarantees of consistency, which are missing from previous approaches like LIME. Bussmann et al. [20] use SHAP to interpretability models for credit risk, showing that models can meet the EU Act requirements for explainability when using the SHAP approach. During a crisis, finding that features related to volatility are the key drivers of attribution, whereas during a stable regime, features related to momentum are the key drivers — and this is a benchmark that is directly relevant to the results that this project expects to find.

To formalize these workflows, Khan et al. [27] did a systematic review and found that explainability methods that are model agnostic are becoming essential for risk validation frameworks for building institutional trust. Likewise, Yeo et al. [28] noted that multi-modal financial structures need to be accompanied with comprehensive post-hoc explainability layers to ensure that the "black box" decision does not conceal data-leakage biases. On this, Liu et al. [29] showed that adding multi-feature inputs and a clear Explainable AI (XAI) framework enables practitioners to visually detect which localized feature is responsible for causing a shift in the state forecast of a model.

2.1.6 Large Language Models in Finance: 2024–2026 Advances

In the last two years, financial analysis using LLM has made significant progress. Developed from a financial corpus of 363 billion tokens, BloombergGPT offers better performance across financial NLP tasks than general purpose LLMs. However, since this is a closed application, it requires the use of GPUs and FinBERT is the suitable solution. In a comprehensive study of financial sentiment analysis models up to Q1 2025, Yang et al. [15] test seven LLMs and conclude that instruction-tuned models outperform base models overall but that FinBERT maintains a good accuracy-to-compute ratio for batch processing of headlines. To bridge the gap between heavy commercial systems and open-source constraints, Bhatia et al. [25] introduced FinTral, a family of multi-modal financial LLMs engineered to match GPT-4 level domain performance while utilizing highly optimized parameter footprints. In real-time financial report analysis, Zhang et al. [19] introduce a retrieval-augmented generation (RAG) framework to predict the sentiment of structured financial text, which further validates the necessity of transformers for sentiment extraction in structured financial text tasks.

2.2 Comparative Analysis of Methodologies

The methodological approaches of the most directly relevant prior studies are summarised below:

Study	Method	Signals Used	Explainability	Lead-Lag Analysis
Ardia et al. [4]	Markov-switching	Price only	No	No
Bollen et al. [24]	Granger causality	Sentiment only	No	Yes (3-day lag)
Riso & Vacca [18]	GARCH + NLP fusion	Price + earnings text	No	Partial
Bussmann et al. [20]	XAI Framework	Credit Risk features	Yes	No
Lundberg et al. [21]	Tree / Post-hoc SHAP	Structural arrays	Yes	No
This project	HMM + GARCH + FinBERT + SHAP	Price + news headlines	Yes	Yes (explicit)

This comparison shows that none of the preceding works have been single studies that integrate all four capabilities: multimodal signal integration, HMM-based regime detection, SHAP explainability, and explicit lead-lag quantification. The present project fills this gap.

2.3 Research Trends and Challenges

The research trends and challenges are described as follows:

- The last few years have seen three major tendencies in the literature: Before 2020, most studies were conducted with lexicon-based models. After 2022, almost all publications employ BERT-family models, both for their enhanced accuracy and reduced costs of inference.
- There is an increasing focus on model interpretability: EU AI Act and SR 11-7 regulatory pressure has led to wider uptake of post-hoc explainability tools. Currently, in 2024-2025, about 35% of applied financial ML papers are using SHAP and LIME.
- Single-modality models are being seen as inadequate for systemic risk detection and multimodal integration is emerging as the frontier. There is a clear trend of joiners in the literature for 2024-2025, where papers are published that integrate structured and unstructured data.

The main ongoing problems are:

(1) Discrepancies between the time when news is published and the time when prices are observed on trading days;

(2) Survivorship bias in historical news datasets because the boundaries of crises are labeled ex post, based on market indicators, not on prospective indicators;

(3) The ambiguity of the crisis labels, as boundaries between crises are not objective but are labeled ex post, based on market indicators, not on prospective ones.

2.4 Research Gap Analysis

2.4.1 Existing Limitations in Prior Work

- Price-only HMM (Hamilton [1] and Shu & Zhu [7]) do not include language signals and thus are not used to check the leading indicator property.

- Sentiment-only studies (Tetlock [17] and Bollen et al. [24]) examine the daily return predictability but do not establish a connection between sentiment and formal regime state changes.
- Multimodal fusion papers (Riso and Vacca [18], Zhang et al. [19]) don't perform an explainability analysis, and function at the return-prediction level instead of the regime-detection level.
- To the best of our knowledge, no previous study has used a multimodal HMM framework to apply SHAP feature attribution to determine what type of signal (price or sentiment) brings about a regime transition.
- None of the studies included in the review showed a comparison with several previous crises that were validated in advance with explicit quantification of the lead time.

2.4.2 Opportunities for Improvement

- Hamilton [1] is a logical extension of our work, where FinBERT sentiment scores are incorporated as extra emission distributions to the HMM.
- The lead-lag cross-correlation between Financial Stress Index and sentiment fear index gives quantitative evidence on the timing of the signals.
- We will show that combining price and sentiment features into a single vector with SHAP allows us to distinguish the strongest sentiment or price signal during each of the historical crises, providing actionable insights for practitioners.

2.4.3 Project Contribution

This project brings a validated, open-source, multi-modal framework that (1) reproduces the work of Wang et al. [31] as a quantitative baseline, (2) integrates sentiment signals from FinBERT, (3) measures sentiment lead-time for three crises and (4) offers explainability at the regime transition level through SHAP analysis. This group of contributions is unique compared to the literature reviewed and immediately applicable for quantitative risk managers looking for signals other than price data to provide early warning.

3. Data Source Identification

3.1 Primary Datasets

Dataset	Source	Pipeline	Format	Size Est.	Period
S&P 500 Daily OHLCV	Yahoo Finance (yfinance)	Person A	CSV via API	< 5 MB	1990–2024
CBOE VIX Index Daily	Yahoo Finance (yfinance)	Person A	CSV via API	< 2 MB	1990–2024
FRED Macro Indicators	Federal Reserve FRED API	Person A	JSON/CSV	< 10 MB	1990–2024
Financial News Headlines	Kaggle: Financial News & Stock Price Integration	Person B	CSV	< 200 MB	2008–2016
FinBERT Pretrained Weights	HuggingFace (ProsusAI/finbert)	Person B	PyTorch	~440 MB (runtime)	N/A
Individual Stock OHLCV (AAPL, JPM, XOM)	Yahoo Finance (yfinance)	Person A	CSV via API	< 15 MB	1990–2024

In addition to the S&P 500 index, three individual stocks are selected across different market sectors to validate the generalizability of the proposed framework: AAPL (Technology), JPM (Financials), and XOM (Energy), all retrieved via yfinance using the same preprocessing pipeline as the index-level data. This cross-sector evaluation addresses the performance of the model beyond aggregate market signals, enabling comparison of regime transition timing and sentiment lead time across distinct economic sectors.

3.2 Backup Datasets

Dataset	Source	Trigger Condition	Size Est.
Daily Financial News for 6000+ Stocks	Kaggle (Genious dataset)	Primary news dataset insufficient coverage	< 300 MB
S&P 500 Historical Data (alternative)	Alpha Vantage API	yfinance downtime or data gaps	< 5 MB
FRED Financial Stress Index (STLFSI)	FRED API	FSI construction validation	< 1 MB
News API Historical Articles	NewsAPI.org (free tier)	Additional crisis-period headline coverage	< 50 MB

3.3 Dataset Descriptions

3.3.1 S&P 500 Daily OHLCV (Yahoo Finance)

Programmatically retrieved from yfinance's Python library (ticker: ^GSPC). The data set includes daily data for open, high, low, close and volume from January 1990 to December 2024, resulting in about 8,800 rows of data for each trading day. No authentication is needed. The data is organized in a Pandas DataFrame with a DatetimeIndex.

Preprocessing requirements: forward filling of rarely missing days; calculation of log returns (log(P_t / P_{t-1})); 21 day rolling annualised volatility; and maximum drawdown over 63 days.

3.3.2 CBOE VIX Index (Yahoo Finance)

The daily closing price of VIX (^VIX) from the same time frame 1990-2024. VIX is utilized as an explicit market implied volatility signal, and also as a part of the Financial Stress Index (FSI). The time series is completely aligned with S&P 500 trading days, bringing no alignment problems.

3.3.3 FRED Macroeconomic Indicators

Data was retrieved from the fredapi Python library, with the series identifiers: FEDFUNDS (Federal Funds Rate), T10Y2Y (10Y-2Y Treasury spread), and BAMLH0A0HYM2 (High Yield OAS spread). These monthly or daily series are interpolated to the frequency of trading days by means of cubic splines, and are auxiliary inputs to the FSI construction.

3.3.4 Financial News Headlines (Kaggle)

The first NLP dataset is the Kaggle "Financial News and Stock Price Integration Dataset" which consists of around 200,000 financial news headlines from Reuters and other financial media covering headlines from different financial institutions by date. Headlines are downloaded in CSV format in these three fields: date, headline text, and (optional) ticker symbol. The data has been collected over a period of 2008-2016 including the period of the GFC recovery and initial volatility. Preprocessing requirements: Deduplication by exact text match; Relaxing the publication date to the closest trading date; Aggregation of all headlines into a single list per trading day for batched FinBERT inference; Storage of mean/max/min sentiment scores grouped by trading day to a CSV file for downstream lead-lag analysis.

3.4 Data Acquisition Methods

- S&P 500 and VIX: yf.download(ticker, start="1990-01-01", end="2024-12-31", auto_adjust=True) called once per session; results cached to CSV.
- FRED data: fredapi.Using the Fred(api_key='FREE_TIER_KEY') object to retrieve a series, serial number 120 requests per minute free tier, and results are saved to a csv file. Data from Kaggle was downloaded using the kaggle datasets download command from the CLI and extracted to Google Colab /content/ folder.
- The weights of FinBERT are loaded at runtime from a transformer.Model: AutoModelForSequenceClassification.from_pretrained('ProsusAI/finbert'); was not persisted to disk.

3.5 Ethical and Licensing Considerations

Data sets are from public APIs or permissively licensed Kaggle repositories. All data shown on Yahoo Finance is pulled from the yfinance open source library and is public market data. The data is licensed as public domain data. Data from the Kaggle news dataset is in the public domain, as per the CC0 license. The weights for FinBERT are licensed under the Apache 2.0 licence, which allows for unrestricted research use. There is no information in any of the data sets that can be used to identify a particular person. There is no commercial use of raw data intended.

3.6 Preliminary Data Assessment

An initial integrity evaluation confirms that the S&P 500 and VIX time-series data from yfinance are structurally complete with no systemic missing values, requiring only standard forward-filling for non-trading calendar days (weekends and holidays). The primary Kaggle financial news dataset features a known coverage constraint during the post-2016 timeline. To maintain analytical feasibility, our framework mitigates this by programmatically mapping news data to direct trading-day timestamps and utilizing our designated backup data repositories to supplement the 2020 and 2022 validation windows.

4. Preliminary Analytical Framework

4.1 Proposed Analytical Techniques

Technique	Purpose	Input	Output
Log-return computation	Stationarise price series	OHLCV close prices	Daily log returns
ARMA-GARCH(1,1)	Conditional volatility estimation	Log returns	Conditional variance series
FSI construction	Composite stress indicator	VIX, GARCH variance, drawdown, spreads	Daily FSI score [0,1]
3-state HMM	Regime labelling	Log returns, GARCH variance, FSI	Daily regime label {0,1,2}

Technique	Purpose	Input	Output
FinBERT inference	Headline sentiment scoring	News headlines (batches of 32)	P(pos), P(neg), P(neutral) per headline
Sentiment aggregation	Daily fear/panic index	Per-headline scores	Daily mean negative sentiment, panic signal
Lead-lag cross-correlation	Signal timing quantification	FSI series, sentiment fear index	Lag (days) at peak correlation
Multimodal fusion	Combined regime prediction	HMM regime probs + sentiment scores	Fused regime probability vector
SHAP analysis	Feature attribution	Fusion model + feature matrix	SHAP values per feature per regime

4.2 Planned Machine Learning Models

4.2.1 Hidden Markov Model (3-State)

Implemented using hmmlearn.hmm.GaussianHMM with n_components=3; full covariance matrices. It includes the following features: daily log returns, GARCH(1,1) conditional standard deviation, 21-day rolling volatility and FSI score. States map to: Stable/Bull = State 0, Volatile/Transitional = State 1, Crisis/Bear = State 2. Multiple random initialisations (n_iter = 100, n_init = 20) are carried out and the one with the maximum log-likelihood is kept. BIC is used for model selection if the data comes in from two to four states. The replication target used by Wang et al. [31] will be used to make the feature engineering the same for the S&P 500 data to ensure they can easily be compared with each other in terms of regime detection performance.

4.2.2 ARMA-GARCH Volatility Model

It is implemented with arch Python Library. The time series consists of ARMA(1,1) model with Student-t innovations. Maximum likelihood estimation is based on rolling 5-year windows, to enable time-varying volatility dynamics. The HMM feature matrix gets conditioned with conditional variance forecasts.

4.2.3 FinBERT Sentiment Classifier

Transformers from HuggingFace were used in the loading of ProsusAI/finbert. Headlines are tokenized by AutoTokenizer(max_length=128, truncation=True) and fed in batch mode with the size of 32. The output probabilities of the softmax, namely {P(positive), P(negative), P(neutral)} are saved for each headline. Daily aggregates: mean P(negative) is the fear index, indicator for days with P(negative) > 40% is the panic signal. Inference is stored in a CSV file for the first time, so as to not recompute it again.

4.2.4 Multimodal Fusion Model

A late fusion approach is a combination of three features : (a) HMM posterior state probabilities [P(S0), P(S1), P(S2)]; (b) daily FinBERT fear index; (c) 3-day and 7-day rolling mean sentiment. A logistic regression classifier (L2 regularization, scikit-learn) is trained on labelled crisis windows (positive class: HMM state 2 onset within the next 5 trading days), and the fused feature vector is given as the input. This is a light version that can still be used with CPU limits in Colab and can be used for interpretable SHAP analysis.

4.3 Evaluation Metrics

Component	Metric	Target / Benchmark
HMM model selection	BIC / AIC	Minimum BIC across 2–4 state configurations
HMM fit quality	Log-likelihood	Stable convergence across 20 initialisations
GARCH model fit	Standardised residual diagnostics	Ljung-Box p > 0.05; Jarque-Bera p > 0.05
FSI validity	Pearson correlation with NBER recession dates	r > 0.6 during crisis windows
Sentiment lead-time	Cross-correlation lag at peak r	Positive lag (sentiment precedes price)

Component	Metric	Target / Benchmark
Fusion model	Precision, Recall, F1 (crisis class)	F1 > 0.70 on held-out crisis windows
SHAP	Feature importance ranking	Consistent with economic intuition per crisis

4.4 Visualization Tools and Planned Charts

- Overlay of the price series of the S&P 500 over a timeline of the regime, using matplotlib to shade the background by HMM state (green=stable, yellow=volatile, red=crisis
- Sentiment trend chart: Daily FinBERT fear index as a time series that is superimposed by VIX, and which is accompanied by an indication of the dates at which crises occur.
- FSI evolution chart: Composite FSI score over time and annotated with crisis event windows.
- Plot of the leads and lags of cross-correlation of FSI and the sentiment fear index for the interval from −30 to +30 days, with the confidence intervals.
- SHAP bar plots: The mean absolute SHAP value per feature is shown per regime state (one plot per state) using the shap library. Confusion model predictions vs model labels for the three validation events (heat map).

4.5 Validation Strategy

The validation is based on events instead of random-split cross validation due to the non-stationary and crisis-specific nature of the prediction target. There are three crisis windows that are held out:

- **2008 GFC**: September 2008 – March 2009 (Lehman collapse through market trough)
- **COVID-19 crash**: February 19, 2020 – March 23, 2020 (peak-to-trough)
- **2022 inflation shock**: January 2022 – October 2022 (Fed tightening cycle onset)

The framework should be able to do the following for each event: (1) assign HMM State 2 within 10 trading days of the onset of the crisis; (2) have a sentiment panic signal that precedes the HMM State 2 assignment; and (3) generate SHAP attributions that are consistent with the economic narrative of each crisis (e.g., dominant credit spread features in 2008, dominant exogenous shock features in 2020).

4.6 Implementation Workflow

Milestone	Week	Deliverable	Owner
M3: Data Pipeline	Wks 1–2	Cleaned price + news CSVs; FSI construction; FinBERT cache	Both
M4: Quantitative Pipeline	Wks 3–4	GARCH estimates; HMM fitted; regime labels; FSI validated	Person A
M4: NLP Pipeline	Wks 3–4	FinBERT scores; fear/panic indices; headline coverage verified	Person B
M5: Lead-Lag Analysis	Wks 5–6	Cross-correlation results; lead-time quantification per crisis	Both
M5: Fusion Model	Wks 5–6	Logistic regression fusion; SHAP values; F1 evaluation	Both
M6: Final Report	Wks 7–8	Full written report; visualisations; reproducible Colab notebook	Both

5. Feasibility and Constraints Assessment

5.1 Technical Feasibility

The project can be accomplished within the stated parameters. No GPU training is required for any model, and they are either pre-trained (FinBERT) or lightweight statistical (HMM, GARCH). Based on Yang et al. [15] benchmarks for similar dataset sizes, the entire pipeline – from data ingestion, to estimating GARCH parameters, fitting HMM, running FinBERT batch inference, computing lead-lag analysis, to computing SHAP – is expected to take less than 45 minutes on Google Colab's CPU runtime.

5.2 Hardware and Software Limitations

Constraint	Impact	Mitigation
Colab CPU-only runtime	FinBERT inference: ~2 sec/batch of 32 headlines	Batch processing + CSV caching after first run
Colab session timeout (12h)	Long-running inference may be interrupted	Checkpoint sentiment CSV every 5,000 headlines
RAM limit (~12 GB free tier)	Large HuggingFace model in memory	Load FinBERT only during NLP pipeline; unload after
Disk quota (~100 GB)	All datasets < 1 GB; well within limits	No action required
Internet dependency for APIs	yfinance / FRED access required	Cache all API results to CSV on first run

5.3 Time Management Considerations

There are no critical path dependencies as the parallel pipeline structure is in place where Person A develops the quantitative modelling and Person B develops the NLP. To avoid dependence of the fusion work in Milestone 5 on the perfect synchronisation of the upstream pipelines, a shared interface CSV (date, regime_label, FSI_score, sentiment_fear_index, panic_signal) is defined in Milestone 3 and managed in terms of integration risk.

5.4 Potential Implementation Challenges

- There is a time gap in news coverage from the primary Kaggle data (2008–2016). In the 2020 and 2022 crisis periods, there may be too few headlines, so the backup Kaggle dataset and/or NewsAPI supplement will be required.

- Label ambiguity in the regime classification: Many HMM state assignments do not clearly correspond to one of the canonical crisis/stable regime classes due to the need for manual examination of the state and relabelling.
- If the cross-correlation peak is at lag zero, then the null result is a valid one and should be carefully framed in the final report.
- GARCH convergence on crisis sub-samples: March 2020 may present a numerical instability with extreme return observations, as there is a fallback on the rolling standard deviation, which is planned in advance.

5.5 Risk Mitigation Summary

Risk	Likelihood	Mitigation Strategy
News dates misaligned with trading days	Medium	Forward-fill to next trading day; neutral score (0.0) for gaps
FinBERT slow on Colab CPU	Medium	Batch size 32; cache CSV; use GPU runtime if available
HMM weak regime separation	Medium	Tune n_states 2–4; multiple random initialisations; select max log-L
Sparse news during 2020 / 2022 crises	Medium	Activate backup Kaggle dataset; supplement with NewsAPI
GARCH non-convergence	Low	GARCH(1,1) default; rolling std fallback; document assumption
Null lead-lag result	Low	Scientifically valid; report as contemporaneous signal finding
Scope creep post-M4	Medium	Freeze scope at M4; no new features added in M5–6

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